R^2 value through machine learning Regression methods:

1.MulTIPLE LINEAR REGRESSION (R^2 value) =0.89507

2. SUPPORT VECTOR MACHINE:

S.no	HYPER PARAMETER	Linear (R^2 value)	RBF(NON LINEAR) (R^2 value)	POLY (R^2 value)	Sigmond (R^2 value)	PreComputed (R^2 value)
1	C=10	-0.03964	-0.05680	-0.053667	-0.054719	Since **Pre-computed** needs a square matrix and since the one in hand is non-square 35 X 7, we have to rule out this option. Note: square matrix are like (5 X 5) , (6 X6)… basically both columns and rows count matches
2	C=100	0.10646	-0.05072	-0.019802	-0.030453	
3	C=1000	0.780283	0.006768	0.266163	0.185068	
4	C=10000	0.923998	0.371895	0.812962	0.853531	
5	C=11000	0.923998	0.399852	0.806530	0.840204	
6	C=14000	0.926182	0.464120	0.777382	0.863734	
7	C=15000	0.930124	0.475584	0.763866	0.873377	
8	C=50000	0.930124	0.666078	0.588505	0.088128	
9	C=100000	0.930124	0.708560	0.400210	-0.843374	

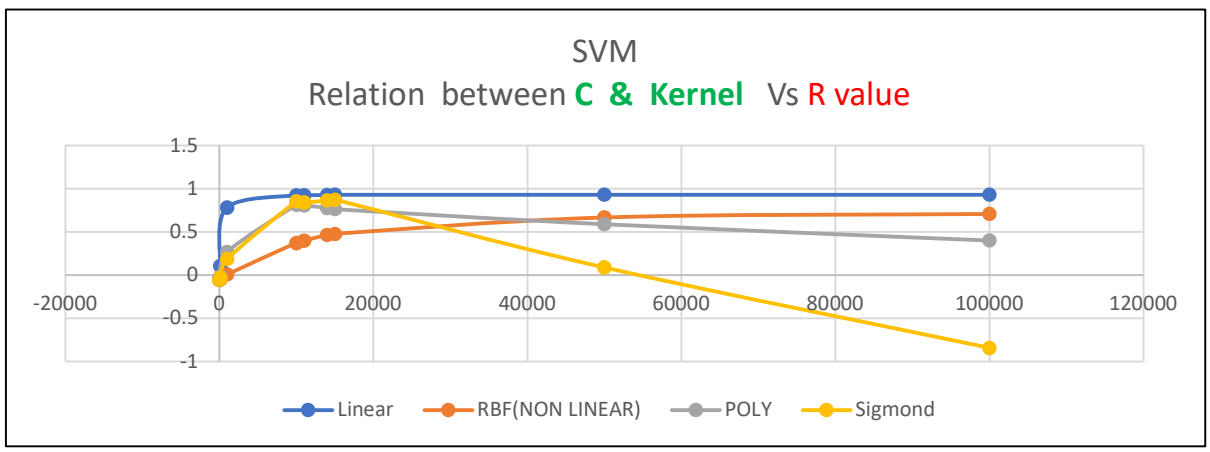
Observation:

1) Kernel=Linear and C=15,000 R^2 value peaks to 93% after that it goes flat.

2) kernel=rbf: shows increasing trend for R^2 as C increases but its peak value of 70% which is still below kernel=Linear.

3)kernel=poly R^2 value peaks at C=14000 and then shows descending trend.

4)Kernel=sigmond R^2 value peaks at C=15000 and then descends drastically to negative value.

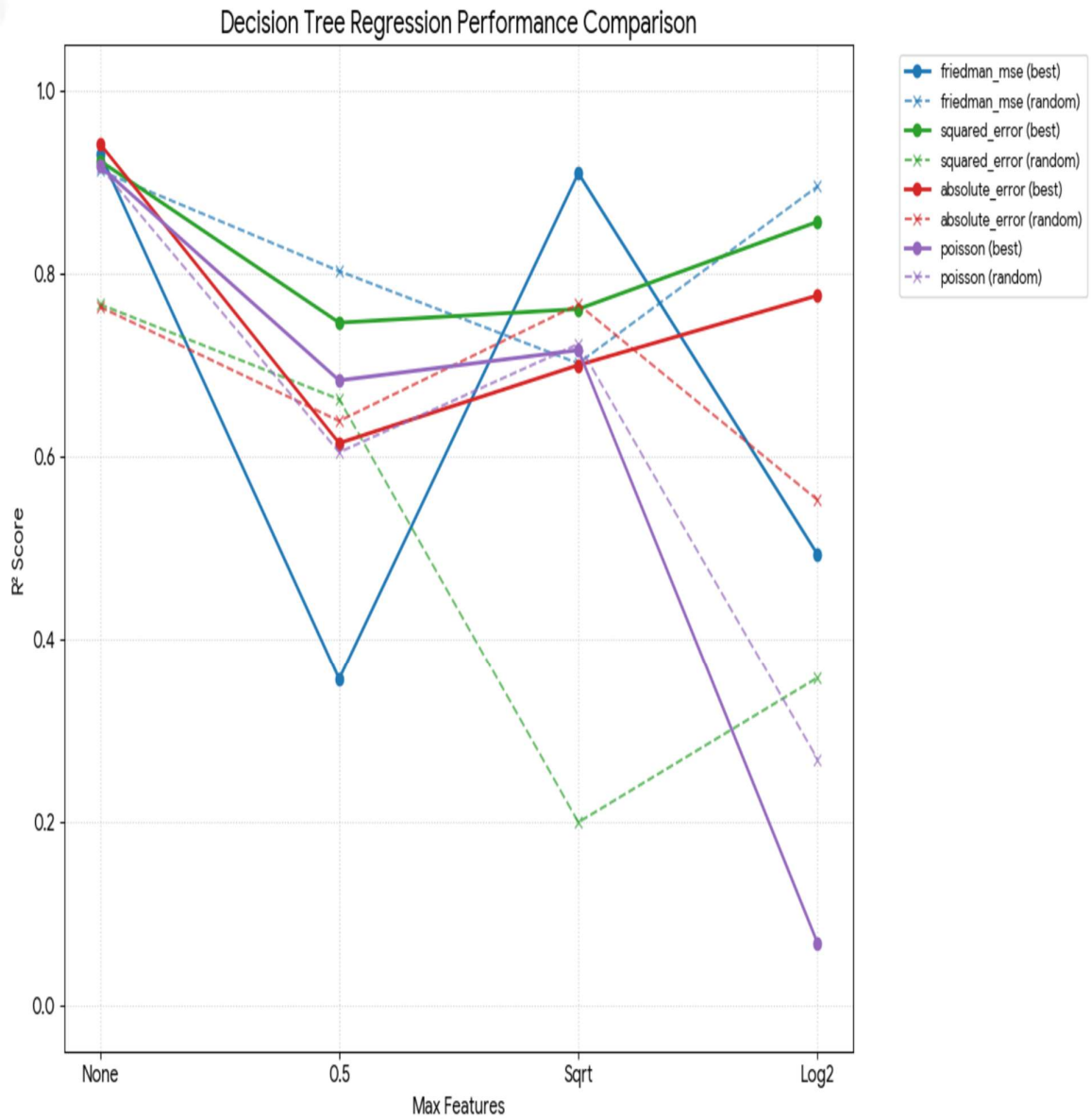


2. Decision Tree:

S.no	CRITERION	MAX_FEATURES	SPLITTER	R^2 value
1	**friedman_mse**	None	best	0.931537
2	**friedman_mse**	None	random	0.912744
3	**friedman_mse**	Int/float 0.5	best	0.356239
4	**friedman_mse**	Int/float 0.5	random	0.802825
5	**friedman_mse**	Sqrt	best	0.909946
6	**friedman_mse**	Sqrt	random	0.701924
7	**friedman_mse**	Log2	Best	0.493029
8	**friedman_mse**	Log2	random	0.895477
9	**squared_error**	None	best	0.923474
10	**squared_error**	None	random	0.766542
11	**squared_error**	Int/float 0.5	best	0.746476
12	**squared_error**	Int/float 0.5	random	0.662607
13	**squared_error**	Sqrt	best	0.761337
14	**squared_error**	Sqrt	random	0.199645
15	**squared_error**	Log2	Best	0.85682
16	**squared_error**	Log2	random	0.357317
17	**absolute_error**	None	best	0.94208
18	**absolute_error**	None	random	0.7637
19	**absolute_error**	Int/float 0.5	best	0.614741
20	**absolute_error**	Int/float 0.5	random	0.638914
21	**absolute_error**	Sqrt	best	0.700024
22	**absolute_error**	Sqrt	random	0.766468
23	**absolute_error**	Log2	Best	0.776166
24	**absolute_error**	Log2	random	0.552544
25	**poisson**	None	best	0.91758
26	**poisson**	None	random	0.9171364
27	**poisson**	Int/float 0.5	best	0.683308
28	**poisson**	Int/float 0.5	random	0.604681
29	**poisson**	Sqrt	best	0.716465
30	**poisson**	Sqrt	random	0.723058
31	**poisson**	Log2	Best	0.068028
32	**poisson**	Log2	random	0.268479

The Decision Tree with best R^2 value:

for CRITERION=absolute_error MAX_FEATURES=None Splitter=best has 0.94208 **R^2 value**


Decision Tree Regression Performance Comparison
1.0
0.8
0.6
0.4
0.2
0.0
R² Score
None
0.5
Sqrt
Log2
Max Features
friedman_mse (best)
friedman_mse (random)
squared_error (best)
squared_error (random)
absolute_error (best)
absolute_error (random)
poisson (best)
poisson (random)

3. RandomForest:

N_estimators=50 random_state=42

S.no	MAX_FEATURES	squared_error	*absolute_error*	friedman_mse	Poisson
1	**None**	0.929063	0.933088	0.930568	0.926729
2	**Int_r_float_0.5**	0.787577	0.775862	0.788161	0.671233
3	**Sqrt**	0.787577	0.775862	0.788161	0.671233
4	**Log2**	0.787577	0.775862	0.788161	0.671233

Important Observation:

When random_state is fixed at 42
for a given CRITERION(say absolute_error) the R2 value generated for all the MAX_FEATURES(int/float,sqrt,log2) gave the same results ;
Except for MAX_FEATURE NONE

N_estimators=50 random_state=none

S.no	MAX_FEATURES	squared_error	*absolute_error*	friedman_mse	Poisson
1	**None**	0.923017	0.946086	0.945538	0.937266
2	**Int_r_float_0.5**	0.810048	0.790675	0.827677	0.778706
3	**Sqrt**	0.777883	0.769185	0.797543	0.707233
4	**Log2**	0.798364	0.812274	0.789760	0.761582

The RandomForest with best R^2 value:

for CRITERION=absolute_error MAX_FEATURES=None Random_State=42 has 0.933088 **R^2 value**

for CRITERION=absolute_error MAX_FEATURES=None Random_State=None has 0.946086 **R^2 value**